

NLP Project Proposal

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Title: Analysis of Indian Parliamentary Proceedings Transcripts Using NLP

Motivation:

Parliamentary proceedings are mostly released as long PDF documents. These are hard to read quickly. Important information such as who spoke, on which topic, and which ministry is being discussed is buried inside many pages of text.

There is no public tool that performs all these analysis steps together in one place. Recent government initiatives such as [Sansad Bhashini](#) show that automated processing of parliamentary data is becoming important. This motivates us to work in this area.

NLP Tasks / Applications

We plan to build a system that processes English parliamentary session documents such as:

- Budget speeches
- Bill introductions
- Question–Answer sessions
- Debates
- Ministerial statements

The system will automatically:

- Identify the speaker, role, party, and ministry
- Detect topics discussed
- Extract named entities such as people, places, schemes, ministries, bills, and money values
- Classify each paragraph into a ministry
- Generate ministry distribution across the document
- Compute topic-wise, speaker-wise, time-based statistics
- Produce simple visualizations

The goal is to convert long unstructured documents into structured data.

The system will work on one document at a time.

Approach:

We model the problem as a document-processing pipeline.

1. Pre-processing: Convert PDFs to text, remove headers and footers, normalize spacing, detect timestamps, and segment the text into speaker turns. Rule-based methods and RegEx will be used as initial baselines.
2. Speaker Identification: We will use rule-based patterns to identify speaker names and roles.
3. Named Entity Recognition: Detect entities such as ministries, schemes, organizations, locations, and money values using pre-trained English NER models.
4. Ministry Classification (Main Task): We treat ministry detection as a paragraph classification problem. We will compare the following approaches:
 - a. Baseline:
Keyword-based scoring using ministry-specific vocabularies.
 - b. Main Method: Embedding-based similarity.
Each ministry's official description will be converted into a semantic embedding. Each paragraph will also be embedded. Cosine similarity will be used to assign the most relevant ministry.
 - c. Optional (if time permits):
Fine-tuning a transformer-based classifier on a small manually labeled dataset.
5. Output: Generate structured output and visual charts

Data:

We will use the [Parliament Digital Library \(PDL\)](#) as our primary source .

We will collect Lok Sabha and Rajya Sabha proceedings, including Budget sessions, debates, and Question Hour transcripts.

References:

- [1] Rohit, Sakala & Singh, Navjyoti. (2018). Analysis of Speeches in Indian Parliamentary Debates. [10.48550/arXiv.1808.06834](#).
- [2] Wijeratne, Yudhanjaya & de Silva, Nisansa & Shanmugarasa, Yashothara. (2019). Natural Language Processing for Government: Problems and Potential. [Link](#)
- [3]Katre, Paritosh. (2019). NLP Based Text Analytics and Visualization of Political Speeches. International Journal of Recent Technology and Engineering. 8. 8574-8579. [10.35940/ijrte.C6503.098319](#).